

PetroChina · KUNLUN Lubricant | TRANSFORMER OIL

KUNLUN KI 25X

Naphthenic insulating oil — Inhibited (contains metal passivator)

PRODUCT DATA SHEET

Product description

KI 25X Transformer Oil is obtained by adding a high-quality antioxidant composite additive to moderately-refined, low-viscosity, low-pour-point, low-sulphur-content naphthenic base oil. It offers rapid heat transfer, good oxidation stability and excellent electrical properties. The product has stable and reliable properties and is produced under a strict quality control procedure.

Quality grade and standards

Quality grade: Highest grade.

Quality indexes according to GB 2536-2011 / IEC 60296:2003. Meets the I-20 °C specification (special purpose).

Application range

Suitable for distribution transformers, generation transformers, power transformers and reactors with a Lowest Cold Start Energizing Temperature (LCSET) of -30 °C, and oil-filled electrical equipment with similar requirements. The product contains a metal passivator.

Properties & characteristics

- Relatively low viscosity and viscosity index for effective cooling and heat transfer
- Excellent heat and oxidation stability to prevent oil ageing and prolong equipment life
- Excellent electrical insulation: high breakdown voltage and low dielectric loss factor
- Good low-temperature properties; free of pour-point depressant; pour point below -30 °C
- Free of DBDS; passes harsh sulphur-corrosion tests (ASTM D1275B, IEC 62535, DIN 51353)
- Environmentally friendly; free of polychlorinated biphenyls (PCB)

Typical properties

Quality indexes and typical data. Quality indexes according to GB 2536-2011 / IEC 60296:2003. Meets the I-20 °C specification (special purpose).

Property	Unit	Method	Min.	Max.	Typical data
Function characteristics					
Pour point	°C	ISO 3016		-30	-33
Kinematic viscosity (40 °C)	mm ² /s	ISO 3104		12	9.8
Kinematic viscosity (-20 °C)	mm ² /s	ISO 3104		1800	500
Water content	mg/kg	IEC 60814		30	<20
Breakdown voltage – untreated	kV	IEC 60156	30		40–60
Breakdown voltage – treated	kV	IEC 60156	70		>70
Density (20 °C)	kg/m ³	ISO 12185		895	882
Aniline point	°C	ASTM D611			79
Dielectric loss factor (90 °C)		IEC 60247		0.005	<0.001
Refining / stability characteristics					
Appearance		Visual		Clear & transparent	Clear & transparent

Property	Unit	Method	Min.	Max.	Typical data
Acid number	mgKO H/g	IEC 62021-1		0.01	<0.01
Interfacial tension	mN/m	GB/T 6541	40		45
Total sulphur content	%	ASTM D5453			<0.01
Corrosive sulphur		ASTM D1275B		Not corrosive	Not corrosive
Corrosive sulphur		DIN 51353		Not corrosive	Not corrosive
Antioxidant additive content	%	IEC 60666		0.3	0.3
Water-soluble acid or base		GB/T 259		Null	Null
2-Furfural content	mg/kg	IEC 61198		0.05	0.02
Operation characteristics – Oxidation stability (120 °C, 500 h)					
Total acid number	mgKO H/g	IEC 61125C		0.3	0.05
Oil sludge	%	IEC 61125C		0.05	0.02
Dielectric loss factor (90 °C)		IEC 60247		0.050	0.010
Gassing property	mm ³ / min	IEC 60628A			+27
Charging tendency	μC/m ³	DL/T 385			1
Health, safety and environment					
Flash point (closed)	°C	ISO 2719	135		142
PCA content	%	IP 346		3	<3
PCB content	mg/kg	IEC 61619		Not detectabl e	Not detectable

Note: Typical data correspond to current production and are provided for reference; they are not a specification limit. The product is produced under a strict quality control procedure with constant product resources. Transformer oil is vulnerable to moisture and contamination – keep dry and clean, store in dedicated containers indoors or in a controlled climate, and keep free of acetylene gas.